

DOCKHUB · DESIGN DECISION RECORD

DockHub recommendations accepted

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DOCUMENT CONTROL

Document ID	DKH-DDR-002	Version	0.1
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/dockhub		
Source	docs/decisions/0002-recommendations-accepted.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	Record Amish's acceptance of all review recommendations, what changed in the repo and the items still open

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted by Amish on 2026-09-25 for every item that carried a recommendation. Items with no recommendation remain proposed, awaiting Amish.

■ Context

DKH-DDR-001 and the review note (docs/REVIEW.md) listed the TRL 2 review items and the TRL 3 engineering proposals. Items with a recommendation had been adopted for TRL 3 work but stayed open for Amish's review. On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." Every item that carried a recommendation is therefore decided as recommended. Where the recommendation named one option among several, that option is the decision. Items with no recommendation are not decided by this record.

■ Decision

Table 1. Items decided by Amish, 2026-09-25: go with recommendation.

#	Item (DKH-DDR-001 number)	Decision	What changed in the repo
1	Budget, R16 (1)	Option (a): build the four-bay cabinet and fit bays 1 and 2 for the first prototype. <code>budget_usd</code> stays at \$1,200 and covers that fit; SwapCell packs are excluded	Already applied at TRL 3 (<code>project.yaml</code> , DKH-REQ-001 R16, <code>bom/bom.csv</code> , <code>cad/src/model.py</code> , DKH-DWG-001). Status wording updated in DKH-DDR-001 v0.2, DKH-REQ-001 v0.4 and <code>bom/bom-notes.md</code> . Budget unchanged: \$1,200 before and after
2	Solar target, R6 (2)	Keep the 400 W canopy as shade and supplement; R6 relaxed from 20 % to 10 %	Already applied in DKH-REQ-001 R6; wording updated in DKH-REQ-001 v0.4 and DKH-PRC-001 v0.4. R6 is still not met (9.3 %)
3	Bay layout (3)	Four bays in one row at hand height	No change; already in DKH-PRC-001 and <code>model.py</code>
4	Charging architecture (4)	One certified charger and one CAN channel per bay	No change; already in DKH-PRC-001 and BOM lines 5 and 6
5	Fire separation (5)	Separate steel compartment per bay with a rear plenum venting through the roof	No change; already in DKH-PRC-001 and <code>model.py</code>
6	Doors during a fire alarm (6)	Bay doors stay locked	No change; already in DKH-PRC-001
7	Solar connection (7)	MPPT controller feeding bay 1, no hybrid inverter	No change to the design; DKH-CAL-001 v0.2 now cites the MPPT control rule as decided
8	Access (8)	Offline-first NFC access with an optional app	No change; already in DKH-PRC-001
9	Pack health gate (9)	Hold packs below 70 % state of health	No change; already in DKH-PRC-001
10	SwapCell interface (12)	Build to SwapCell interface v0.3	No change; already cited in DKH-REQ-001 R2 and DKH-PRC-001
11	TRL 3 engineering proposals (13)	Release packs at 80 % state of charge or more; chargers with a power factor of 0.9 or better; plinth flush with the body (1,000 x 500 mm); 15 min rider patience and the demand profile of DKH-CAL-001	No change to numbers; DKH-CAL-001 v0.2 and <code>sizing.py</code> now cite the release threshold as decided
12	Two-bay first prototype that cannot run the duty case (review note, TRL 3 session)	Accepted with item 1: the prototype demonstrates the station with one resident pack; the duty case needs the four-bay fit (\$1,423)	Recorded here and in <code>docs/REVIEW.md</code>

No design change follows from these decisions: each was already built into the TRL 3 model, BOM, drawing and calculations. The GA drawing DKH-DWG-001 stays at Rev P1 (regenerated, geometry and notes unchanged), and DKH-CAL-001 results are unchanged.

The TRL 2 review made no recommendation to reword the `project.yaml` pitch or problem, so they are unchanged.

■ Items still open

Table 2. Proposed, awaiting Amish (no recommendation was made).

Item	Options
Pack ownership model (DKH-DDR-001 item 10)	Station-owned pool, rider-owned one-for-one exchange, or fleet packs; to follow co-design
First pilot site and partner (DKH-DDR-001 item 11)	A delivery-worker group in a high-income city, or a moto-taxi hub in East Africa
How to close R6 (9.3 % against 10 %)	Accept about 9 %, route flat packs to bay 1 in daylight and accept occasional waits, feed more bays from the sun, or drop the target
How to close R9 (charging pauses above about 29 to 39 °C ambient)	Accept daytime pauses in hot climates, add active cooling, narrow the design ambient, or raise the charge temperature limit with SwapCell

■ Consequences

- Requirement status is unchanged: R6 and R9 not met; R8, R15 and R16 at risk; R7 and R14 not verifiable at TRL 3; nine met.
- Cross-repo actions (recorded in `docs/REVIEW.md`, not changed here): ask SwapCell for the pack wake time on the coded INTERLOCK loop, which sets the 2 s return check; note for SwapCell that multi-bay hosts need a CAN channel per bay because packs default to node 0.
- TRL 4 work (bench bay, propagation test, anchor and charger selection with datasheets, purchasing) stays on hold by Amish's instruction. `trl` and `trl_target` stay at 3.